Can a Quantum Circuit Read a Sentence?

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Abstract

Quantum natural language processing (QNLP) rests on a mathematical coincidence. Grammar composes the meaning of a sentence in the same way that qubits compose into a larger quantum state. A sentence can therefore be read as a recipe for a quantum circuit, and the circuit's measurement can be read as the sentence's meaning. We study this idea at the smallest scale where every step can be checked by hand. Our model is a sentiment classifier that maps a short sentence to a two-qubit circuit. Grammar decides which word rotates which qubit, an entangling gate ties the subject to the predicate, and one measurement returns positive or negative. Each prediction is the result of simulating the circuit exactly. The parameters are trained with the parameter-shift rule, the analytic-gradient method also used on quantum hardware. On a small balanced dataset the classifier labels every sentence correctly. We trace one example through the full pipeline, from parse to final amplitude. The goal is understanding rather than performance. The paper gives a working demonstration of how syntax can drive a quantum circuit, and a clear account of what two qubits can and cannot show.

1. Introduction

Most natural language processing learns vector representations of words from large corpora. Grammar, if it appears at all, is something the model discovers on its own. QNLP starts from the other side. In the DisCoCat model of Coecke et al. (2010), short for distributional compositional categorical, grammar comes first. A noun is a state. A verb is a process that consumes the nouns around it. The grammatical reduction of a sentence specifies how to combine those pieces. Read literally, that diagram is a quantum circuit (Coecke et al., 2020). The analogy is more than a slogan because compositional semantics and multi-qubit Hilbert space are both built from tensor products. The bookkeeping is the same on both sides.

We wanted to see this work in practice. We built the smallest model that still does something useful: a binary sentiment classifier for short sentences such as "the clever program runs." It uses two qubits, a closed vocabulary, and three sentence templates. At this size the circuit can be simulated exactly, trained the way a real QNLP circuit would be, and checked line by line.

Two details keep the exercise honest. First, the prediction is read off a simulated quantum state through the Born rule, not off a classical sigmoid placed next to a circuit diagram. Second, the parameters are trained with the parameter-shift rule (Mitarai et al., 2018; Schuld et al., 2019), the exact gradient method required on physical hardware. We are also clear about scale. Two qubits can show compositionality at work. They cannot demonstrate a quantum advantage. Section 6 states where the hard problems begin.

2. Background

2.1. Qubits and Measurement

A classical bit is either 0 or 1. A qubit is a unit vector in $\mathbb{C}^2$, written in the computational basis as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{1}$$

The only single-qubit gate we need is the R_y rotation,

$$R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \tag{2}$$

which sends $|0\rangle$ to $\cos(\theta/2)\,|0\rangle + \sin(\theta/2)\,|1\rangle$. This is a point at polar angle θ below the north pole of the Bloch sphere. Measuring in the $\{|0\rangle, |1\rangle\}$ basis collapses the state onto one pole. The probabilities follow the Born rule,

$$P(|1\rangle) = |\langle 1|\psi\rangle|^2 = \sin^2(\theta/2), \tag{3}$$

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so a state on the equator ($\theta = \pi/2$) is a fair coin. We use only real amplitudes and R_y gates. The whole computation then lives on a single great circle of the Bloch sphere, which is the right level of machinery for a first look at QNLP (Nielsen & Chuang, 2010).

2.2. Two Qubits and Entanglement

Two qubits live in $\mathcal{H}_{\text{subj}} \otimes \mathcal{H}_{\text{sent}} \cong \mathbb{C}^4$, with basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. The left digit is the subject qubit and the right digit the sentence qubit, so amplitude i corresponds to $i = 2\,(\text{subject}) + (\text{sentence})$. A single-qubit gate acts through a Kronecker product, for example $R_y(\theta) \otimes I$ on the subject. The controlled-NOT gate, with the subject as control, flips the sentence qubit when the subject is $|1\rangle$. It swaps $|10\rangle \leftrightarrow |11\rangle$:

$$\text{CX} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{4}$$

When both qubits are already in superposition, this gate produces a state that does not factor as $|\psi_{\text{subj}}\rangle \otimes |\psi_{\text{sent}}\rangle$. The two qubits become entangled. This is the behavior we want from grammatical composition, where subject and predicate can no longer be read in isolation.

2.3. Grammar in DisCoCat

DisCoCat assigns each word a grammatical type built from a noun type n and a sentence type s (Coecke et al., 2010). A transitive verb carries type $n^r\, s\, n^l$. It expects a noun on each side and returns a sentence. Reducing the sentence contracts the matching wires, called cups, and the surviving wire is the sentence meaning. Reading the diagram as a quantum process turns nouns into states, verbs into multi-linear maps, and cups into Bell-style contractions (Coecke et al., 2020). Our model keeps the central idea, that grammar fixes how meanings combine before any number is computed. It reduces the apparatus to two qubits so the whole thing can be simulated and trained exactly.

3. The Model

3.1. Constrained Grammar

To remove any ambiguity, we restrict every sentence to one of three templates over a fixed vocabulary:

the ADJ NOUN VERB	intransitive
the ADJ NOUN VERB the NOUN	transitive
the ADJ NOUN VERB the ADJ NOUN	+ obj. adj.

A small parser reads each sentence into its grammatical roles: a subject noun, a subject adjective, a verb, and optionally an object noun and its adjective. This replaces the CCG and DisCoCat machinery that a library such as lambeq (Kartsaklis et al., 2021) would provide at scale. We give up coverage for a parse we can trust and reproduce. Figure 1 shows one parse.

Table 1. Role weights ω_{role} in the sentiment score of Equation (7). Verbs and subject adjectives carry the sentiment here and are weighted accordingly. Nouns contribute little. Object weights apply only when the sentence has an object.

Role	Weight ω
Verb	1.40
Subject adjective	1.25
Object adjective	0.90
Subject noun	0.45
Object noun	0.35

3.2. Encoding Words as Angles

Each content word w carries one trainable number ϕ_w, initialized near zero. Grammar enters twice. First, it decides which qubit a word turns. The subject-side words, the adjective and subject noun, sum into the subject qubit's rotation. The predicate-side words, the verb and any object or object adjective, sum into the sentence qubit's rotation:

$$\theta_{\text{subj}} = \phi_{\text{adj}} + \phi_{\text{subj}}, \tag{5}$$

$$\theta_{\text{pred}} = \phi_{\text{verb}} + \phi_{\text{obj-adj}} + \phi_{\text{obj}}. \tag{6}$$

Absent roles drop out of the sums.

Second, grammar decides how strongly each role contributes. A role-weighted sum forms a single sentiment score. Verbs and adjectives carry most of the sentiment in this domain, so they receive larger weights than the nouns:

$$s = b + \sum_{(\text{role},\, w) \in \text{roles}} \omega_{\text{role}}\, \phi_w, \tag{7}$$

where the role weights ω appear in Table 1 and b is a trainable bias. The score sets the final rotation, offset by $\pi/2$ so that a neutral sentence leaves the sentence qubit on the equator:

$$\theta_{\text{dec}} = \frac{\pi}{2} + s. \tag{8}$$

3.3. Circuit and Readout

The circuit (Figure 2) runs four steps. It applies $R_y(\theta_{\text{subj}})$ to the subject qubit and $R_y(\theta_{\text{pred}})$ to the

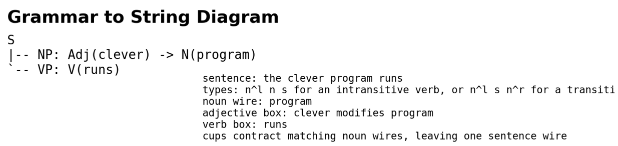

Figure 1. The grammatical parse of an example sentence. These roles, not the order the words happen to arrive in, decide how the angles are wired into the circuit of Figure 2.

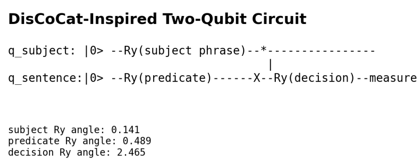

Figure 2. The two-qubit circuit. Word parameters set the opening rotations. The CX gate entangles subject and sentence, which is grammatical composition. A final $R_y(\theta_{\mathrm{dec}})$ writes in the sentiment verdict before the sentence qubit is measured.

sentence qubit to load the words. It applies a CX with the subject as control to compose them. It applies $R_y(\theta_{\mathrm{dec}})$ to the sentence qubit for the verdict. It measures the sentence qubit. Starting from $|00\rangle$, the state after all four gates is

$$|\Psi\rangle = (I \otimes R_y(\theta_{\mathrm{dec}}))\, \mathrm{CX} \times (R_y(\theta_{\mathrm{subj}}) \otimes R_y(\theta_{\mathrm{pred}}))\ |00\rangle\,. \tag{9}$$

The model's only output is the probability of reading 1 on the sentence qubit,

$$P(|1\rangle) = |\Psi_{01}|^2 + |\Psi_{11}|^2. \tag{10}$$

We call $|1\rangle$ positive and $|0\rangle$ negative, and classify by whether $P(|1\rangle)$ exceeds one half. No separate scoring function runs offstage. The circuit we draw and the prediction we make are the same code.

3.4. Training

We train by minimizing binary cross-entropy between $P(|1\rangle)$ and the label. The gradients are the interesting part. Each angle enters through a gate of the form $e^{-i\theta G/2}$. For such gates the derivative of a measured probability is given exactly by the parameter-shift rule (Mitarai et al., 2018; Schuld et al., 2019):

$$\frac{\partial P}{\partial \theta} = \tfrac{1}{2}\left[\, P\left(\theta + \tfrac{\pi}{2}\right) - P\left(\theta - \tfrac{\pi}{2}\right)\right]. \tag{11}$$

Table 2. The statevector for "the clever program runs" at each stage of Equation (9). The left digit is the subject qubit, the right digit the sentence qubit. The $|01\rangle$ and $|11\rangle$ columns are the ones a positive reading draws on.

Operation	$\|00\rangle$	$\|01\rangle$	$\|10\rangle$	$\|11\rangle$
init $\|00\rangle$	1.000	0.000	0.000	0.000
word R_y (both qubits)	0.968	0.241	0.068	0.017
CX: $\|10\rangle \leftrightarrow \|11\rangle$	0.968	0.241	0.017	0.068
decision $R_y(2.465)$	0.094	0.993	−0.059	0.039

This is not a finite-difference approximation. It is exact for these gates, and it is how real QNLP circuits are differentiated on hardware. We evaluate it for each of the three gate angles by re-running the simulation at shifted values. The chain rule then carries those gradients back to the word parameters and the bias. One word may feed several gates, weighted by how often it appears and by its role. Gradient descent does the rest, over fifty epochs at a learning rate of 0.35.

4. Worked Example

To see the model run, take the positive sentence "the clever program runs." After training, its words have settled on $\phi_{\mathrm{clever}} = 0.098$, $\phi_{\mathrm{program}} = 0.043$, and $\phi_{\mathrm{runs}} = 0.489$, with bias $b = 0.068$.

The subject qubit collects the adjective and noun, $\theta_{\mathrm{subj}} = 0.098 + 0.043 = 0.141$. The sentence qubit collects the verb, $\theta_{\mathrm{pred}} = 0.489$. The role-weighted score from Equation (7) is

$$s = 0.068 + 1.25(0.098) + 0.45(0.043) + 1.40(0.489) = 0.894,$$

so the verdict rotation is $\theta_{\mathrm{dec}} = \pi/2 + 0.894 = 2.465$. The verb, at weight 1.40, dominates the score, which is the bias we built in.

Following the four amplitudes of Equation (9) through the circuit gives Table 2. The opening rotations barely disturb $|00\rangle$. The CX swaps the $|10\rangle$ and $|11\rangle$ amplitudes, which is grammatical composition in action. The verdict rotation tips the sentence qubit across the equator, from the $|0\rangle$ hemisphere into the $|1\rangle$ hemisphere. Reading off Equation (10),

$$P(|1\rangle) = 0.993^2 + 0.039^2 = 0.988 \ \Rightarrow\ \text{POSITIVE}.$$

The opposite sentence, "the broken code fails," runs through the same circuit with its own angles and gives $P(|1\rangle) = 0.003$, a confident negative. Almost the entire gap between the two comes from the verb and adjective angles, as the role weights intend.

Table 3. Accuracy. Every prediction is the outcome of simulating and measuring the two-qubit circuit (Equation (10)).

Split	# Sentences	Accuracy
Train	60	100%
Test	12	100%

5. Results

The dataset has 72 sentences, split evenly into 36 positive and 36 negative, with 60 for training and 12 for testing. Positive and negative sentences use identical templates and differ only in their content words, such as clever or reliable against broken or buggy. This sharpens the question. With grammar fixing the wiring, can the learned word angles alone carry the meaning? The splits, the parser, and the random seed are all deterministic, so the run reproduces from a single command.

It can. The model labels every training and test sentence correctly (Table 3). The cross-entropy falls smoothly across the fifty epochs of training (Figure 3). Every prediction comes from exact statevector simulation of the two qubits. At this size a forward pass and its gradients finish in well under a second on a laptop.

Perfect accuracy here is not a strong claim. The dataset is small, grammatically uniform, and separable in the feature the model was built around. A classical linear classifier would also solve it. The experiment checks something narrower. It checks that the quantum realization holds together. Rotations, entanglement, a Born-rule readout, and parameter-shift training reproduce the correct meaning from end to end, with grammar rather than word order doing the composing. The result is a working mechanism, not a benchmark.

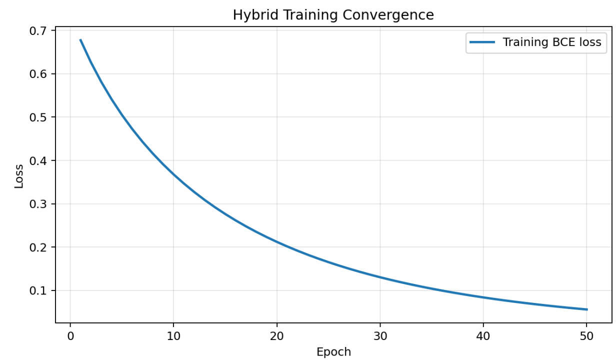

Figure 3. Binary cross-entropy over training. The gradients behind this curve come from the parameter-shift rule of Equation (11), the same method used on a real quantum device.

6. Limitations

The ceiling is worth stating plainly. Two qubits and a fixed grammar cannot demonstrate a quantum advantage, and we do not claim one. Simulating a circuit this small is trivial on a classical computer. The value is the demonstration itself. A sentence's grammar can be compiled into a circuit whose measurement encodes meaning, with no hidden steps.

Real QNLP changes the picture in ways this model does not capture. The hand-written parser becomes a full DisCoCat or CCG pipeline (Kartsaklis et al., 2021). Words spread across several qubits each. The circuits run on noisy devices where probabilities are sampled rather than computed, so the analytic gradients of Equation (11) become a necessity rather than a convenience (Lorenz et al., 2023). New difficulties appear that do not exist at two qubits. The sharpest is the barren plateau, where training gradients shrink exponentially as qubits are added (McClean et al., 2018). This is a central obstacle for variational quantum models in general (Cerezo et al., 2021).

The modeling shortcuts also have a cost. Real amplitudes, one rotation per word, a fixed table of role weights, and a closed vocabulary each trade expressivity for clarity. Richer QNLP encodes words as multi-qubit states and verbs as genuine multi-linear maps, so that composition is more than a single scalar score. Extending the model in that direction, without losing the transparency that makes it useful, is the natural next step.

7. Conclusion

We built a two-qubit sentiment classifier in which grammar lays out the circuit, learned word angles supply the meaning, and one measurement gives the verdict. Training uses the parameter-shift rule throughout. On a small balanced dataset the model is perfectly accurate. We traced a single sentence through every stage, from parse and angles to the entangling gate, the amplitudes, and the readout. The model is small by design. It offers a clear picture of how syntax can drive a quantum circuit, and a clear sense of how much further there is to go.

The dataset, parser, simulator, trainer, and figures are released with the project, and the full pipeline runs deterministically from one command.

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